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# Detection of Ventilatory Thresholds using Near-Infrared Spectroscopy with a Polynomial Regression Model

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## Abstract

Whether near-infrared spectroscopy (NIRS) is a convenient and accurate method of determining first and second ventilatory thresholds ($VT_1$ and $VT_2$) using raw data remains unknown. This study investigated the reliability and validity of $VT_1$ and $VT_2$ determined by NIRS skeletal muscle hemodynamic raw data via a polynomial regression model. A total of 100 male students were recruited and performed maximal cycling exercises while their cardiopulmonary and NIRS muscle hemodynamic data were measured. The criterion validity of $VT_{1VET}$ and $VT_{2VET}$ were determined using a traditional V-slope and ventilatory efficiency. Statistical significance was set at α =. 05. There was high reproducibility of $VT_{1NIRS}$ and $VT_{2NIRS}$ determined by a NIRS polynomial regression model during exercise ($VT_{1NIRS}$, r = 0.94; $VT_{2NIRS}$, r=0.93). There were high correlations of $VT_{1VET}$ vs $VT_{1NIRS}$ (r = 0.93, $p < .05$) and $VT_{2VET}$ vs $VT_{2NIRS}$ (r = 0.94, $p < .05$). The oxygen consumption ($VO_2$) between $VT_{1VET}$ and $VT_{1NIRS}$ or $VT_{2VET}$ and $VT_{2NIRS}$ was not significantly different. NIRS raw data are reliable and valid for determining $VT_1$ and $VT_2$ in healthy males using a polynomial regression model. Skeletal muscle raw oxygenation and deoxygenation status reflects more realistic causes and timing of $VT_1$ and $VT_2$.

**Key words:** ventilatory threshold, anaerobic threshold, oxygen consumption, polynomial regression, NIRS

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## 1. Introduction

Maximal oxygen consumption ($VO_{2max}$) is considered the gold standard for evaluating cardiorespiratory function (Mitchell et al., 1958; Saltin and Astrand, 1967). The submaximal anaerobic threshold (AT) and ventilatory threshold (VT) provide a better clinical index of aerobic fitness than $VO_{2max}$. Previous studies confirmed that the VT was also related to the intensity of aerobic exercise, self-reported activity levels, and training benefits (Tamai et al., 1993; Gaskill et al., 2001). Therefore, developing a convenient and accurate method to determine VT is a significant research issue.

Other methods of estimating the VT have been developed such as tissue deoxygenation (Bhambhani et al., 1997), double product breakpoint (Omiya et al., 2004), non-linear increased EMG activity (Lucia et al., 1997), heart rate deflection (Conconi et al., 1982), and heart rate variability (Cottin et al., 2007). One study indicated that changes in the respiratory frequency were related to the first ventilatory threshold ($VT_1$) and second ventilatory threshold ($VT_2$) (Neder and Stein, 2006). The most popular technique is the V-slope method (Beaver et al., 1986), which determines the VT by measuring the non-linear point of increase in the slope of carbon dioxide production ($VCO_2$) vs oxygen uptake ($VO_2$) during incremental exercise. The VT reflects lactic acid accumulation during exercise that causes hyperventilation. (Myers and Ashley, 1997).

Near-infrared spectroscopy (NIRS) signals obtained during exercise were confirmed to reflect local tissue oxygen delivery, utilization, and blood flow (Boushel et al., 2001; DeLorey et al., 2003). The primary NIRS parameters include oxygenated hemoglobin ($O_2Hb$), oxygenated myoglobin ($O_2Mb$), deoxygenated hemoglobin (HHb),

and deoxygenated myoglobin (HMb). $O_2Hb$ and $O_2Mb$ were confirmed to reflect tissue $O_2$ delivery. HHb and HMb were confirmed to reflect tissue $O_2$ utilization.

The physiological significance of muscle oxygen saturation was demonstrated using different exercise intensities and modes. Previous studies reported that skeletal muscle oxygenation was relatively constant during walking and gradually increased during incremental running (Hiroyuki et al., 2002; Lee et al., 2011). Other studies indicated that muscle oxygenation status follows a sigmoidal profile during incremental cycling (Ferreira et al., 2007; Legrand et al., 2007; Thomas and Stephane, 2008; Racinais et al., 2014; Boone et al., 2015). Studies reported that muscle $O_2$ delivery ($O_2Hb$ + $O_2Mb$) increased until moderate intensity and then decreased or was maintained until exercise ended. Other studies found that $O_2$ delivery decreased from the start to the end of exercise. Further studies indicated that muscle $O_2$ utilization (HHb + HMb) increased from the beginning to the end of exercise. Other studies found that $O_2$ utilization increased to high intensity and were then maintained until the end of exercise.

Previous studies indicated that $VT_1$ and $VT_2$ could be determined using NIRS based on the breakpoint of skeletal muscle oxygenation and blood flow (deoxy[Hb + Mb] and total hemoglobin [Hb + Mb]) (Boone *et al.*, 2015), muscle oxygenation ($O_2Hb$) (Legrand *et al.*, 2007; Racinais *et al.*, 2014), the difference between muscle oxygenation and deoxygenation (Δ[$O_2$HbMb-HHbMb]) (van der Zwaard *et al.*, 2016), and respiratory muscle oxygenation (Moalla *et al.*, 2005). Mathematical models of determining $VT_1$ and $VT_2$ include single linear regression (Legrand *et al.*, 2007), double linear regression (Grassi *et al.*, 1999; Moalla *et al.*, 2005; Legrand *et al.*, 2007; Wang

*et al.*, 2012), non-linear models (Racinais *et al.*, 2014), and combining linear and polynomial regression (van der Zwaard *et al.*, 2016). Integrating past studies, the choice of parameters for determining $VT_1$ and $VT_2$ were muscle oxygenation ($O_2Hb$ + $O_2Mb$) or deoxygenation (HHb + HMb). The muscle oxygenation and deoxygenation data in most mathematical models are polynomial patterns (sigmoidal profile).

Although NIRS was confirmed as a non-invasive method to reflect oxygen delivery, utilization, and blood flow during exercise, whether oxygenation and deoxygenation NIRS signals can be used to determine VT and RCP requires further clarification. Establishing mathematical models using raw data can reduce data conversion errors. Polynomial regression model verification is also used as a reference for future development of NIRS to automatically determine $VT_1$ and $VT_2$. This study investigated the reliability and validity of $VT_1$ and $VT_2$ determined via NIRS oxygenation and deoxygenation raw data using a polynomial regression model. This study's hypotheses were that $VT_1$ and $VT_2$ determined using raw NIRS data and a polynomial regression model would have good reliability and viability.

## 2. Materials and methods

### 2.1 Subjects

This study was approved by the ethics committee of Chang Gung Memorial Hospital. The recruitment criteria for the male college students were 1) non-smokers, 2) no medication or vitamin use, and 3) no cardiopulmonary or hematological diseases. A total of 100 subjects were recruited and all provided informed consent. The subjects were instructed to refrain from exercise for ≥ 24 h before the exercise test. The

subjects arrived at the testing center at 10:00 AM to eliminate any possible diurnal effect. All of the subjects completed the GXT once and 20 repeated the GXT after 48 h to confirm the test's reliability.

**2.2 Graded exercise test (GXT)**

Each subject underwent a GXT to determine $VT_1$ and $VT_2$. The GXT was measured using a bicycle ergometer (Corvial 400, Lode). After 2 min of unloaded pedaling, the loading was increased by 30 Watt every 3 min until exhaustion (Wang *et al.*, 2010; Mao *et al.*, 2011). Heart rate (HR), blood pressure (BP), minute ventilation ($V_E$), oxygen consumption ($VO_2$), and $CO_2$ production ($VCO_2$) were collected using a biomedical system (Powerlab, USA). Arterial $O_2$ saturation was monitored via finger pulse oximetry (model 9500, Nonin Onyx). The $VO_{2max}$ criteria were 1) the level of $VO_2$ increased < 2 $ml \cdot kg^{-1} \cdot min^{-1}$ over ≥ 2 min, 2) HR exceeded the predicted maximal heart rate ($HR_{max}$), and 3) the respiratory exchange ratio (RER) exceeded 1.2 (Wang *et al.*, 2010; Mao *et al.*, 2011).

**2.3 Determination of $VT_1$ and $VT_2$**

$VT_1$ was determined by an increase in $V_E/VO_2$ with no increase in $V_E/VCO_2$ and departure from the linearity of ventilation ($V_E$), whereas $VT_2$ corresponded to an increase in both $V_E/VO_2$ and $V_E/VCO_2$ (Beaver et al., 1986). All of the $VT_1$ and $VT_2$ assessments were confirmed by visually inspecting graphs of the time plotted against each relevant respiratory variable. $VT_1$ and $VT_2$ were independently determined by

two experienced exercise physiologists. When the results diverged, $VT_1$ and $VT_2$ were judged by a third exercise physiologist.

### 2.4 NIRS skeletal muscle hemodynamics

Local skeletal muscle hemodynamics profiles were simultaneously monitored using an NIRS system (Oxymon, Artinis, Netherlands). The transmitting and receiving optode was placed on the left vastus lateralis (VL) muscle (12 cm above the proximal border of the patella and 3-5 cm lateral to the midline of the thigh) (Wang *et al.*, 2010). The distance between the optodes was adjusted to ensure proper placement (range: 2.5-3.5 cm) and good signal strength (10-30%) and the sampling frequency was set at 10 Hz. $O_2Hb$ and HHb were presumed to be reliable estimators of changes in tissue oxygenation or deoxygenation status representing regional $O_2$ delivery and utilization. tHb reflected changes in the tissue blood volume (Boushel *et al.*, 2001; DeLorey *et al.*, 2003).

### 2.5 Determination of $VT_{1NIRS}$ and $VT_{2NIRS}$

The following procedures were used to determine tissue NIRS reflecting $VT_1$ and $VT_2$: 1) 10 Hz $O_2Hb$ and HHb data were plotted from the beginning to the end of GXT; 2) the trend lines were drawn and adjusted using a polynomial regression model with Microsoft Excel; 3) $VT_{1NIRS}$ was determined by the peak of $O_2Hb$ polynomial function (the maximal value of $O_2Hb$); and 4) $VT_{2NIRS}$ was determined by the peak of HHb polynomial function (the maximal value of HHb) (Fig. 1).

**Polynomial regression function** $\mathbf{Y = aX^3 + bX^2 + cX^1 + d}$

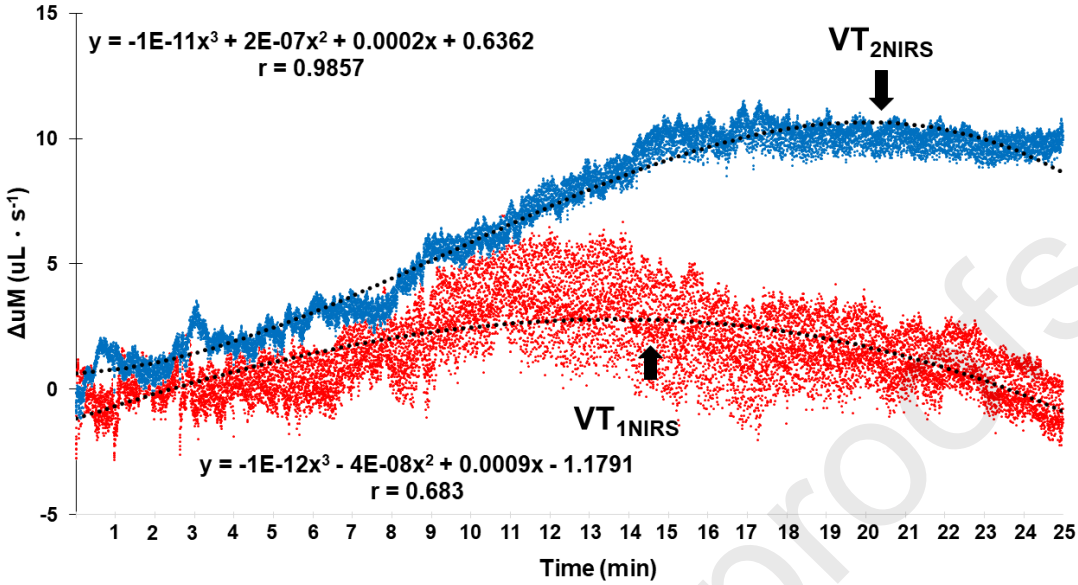


Figure 1. NIRS signal changes during the GXT in one subject. Red dots: oxygenated hemoglobin. Blue dots: deoxygenated hemoglobin.

### 2.6 Statistical analysis

All of the cardiopulmonary data collected during the GXT were calculated at an average of 60 s. If there was an obvious deviation in the data, such as extreme values due to the subject sneezing or coughing, then that data were deleted. If the average exceeded 4 standard deviations, then the value was listed as an extreme value and deleted. The data were analyzed by SPSS for Windows 23.0. The independent t-test was used to compare $VO_2$ at $VT_{1NIRS}$ and $VT_{2NIRS}$ between the first ($T_1$) and second ($T_2$) tests for reliability. The paired t-test was used to compare $VO_2$ between $VT_{1VET}$ vs $VT_{1NIRS}$ and $VT_{2VET}$ vs $VT_{2NIRS}$. Pearson's correlation was used to evaluate the association of $VO_2$ of $VT_{1VET}$ vs $VT_{1NIRS}$ and $VT_{2VET}$ vs $VT_{2NIRS}$. A Bland-Altman analysis (Bland and

Altman, 1986) was used to assess similarities between the ventilatory and NIRS methods. Statistical significance was α = .05.

**3. Results**

The subjects' basic anthropometric and cardiopulmonary characteristics are presented in Table 1. To confirm the reliability of $VT_{1NIRS}$ and $VT_{2NIRS}$ during the GXT between $T_1$ and $T_2$, the correlations and similarities were assessed. There were significant positive correlations in $VT_{1NIRS}$ and $VT_{2NIRS}$ between $T_1$ and $T_2$. The Bland-Altman analysis demonstrated that the mean differences between $VT_{1NIRS}$ and $VT_{2NIRS}$ were all within a 95% (± 1.96 SD) confidence interval (Fig. 2).

There were significant positive correlations between the $VO_2$ of $VT_{1VET}$ vs $VT_{1NIRS}$ and $VT_{2VET}$ vs $VT_{2NIRS}$, validating $VT_1$ and $VT_2$ measured by the ventilatory and NIRS methods. The Bland-Altman analysis showed that the mean differences in $VT_{1VET}$ vs $VT_{1NIRS}$ and $VT_{2VET}$ vs $VT_{2NIRS}$ were consistent (Fig. 3). The comparisons of $VO_2$ between $VT_{1VET}$ vs $VT_{1NIRS}$ and $VT_{2VET}$ vs $VT_{2NIRS}$ demonstrated that there were no significant differences between the ventilatory and NIRS methods (Table 2).

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Table 1 The subjects' basic anthropometric and cardiopulmonary characteristics

| Anthropometric | | |
|---|---|---|
| Age (years) | 22 ± 3 | |
| Weight (kg) | 67.1 ± 10.5 | |
| Height (cm) | 173.1 ± 6.0 | |
| BMI (kg・$m^{-2}$) | 22.3 ± 0.3 | |
| **Cardiopulmonary** | **Baseline** | **Peak exercise** |
| Workload (watt) | 0 ± 0 | 210 ± 5 |
| HR (beat・$min^{-1}$) | 77 ± 3 | 193 ± 2 |
| $V_E$ (L・$min^{-1}$) | 6.8 ± 0.5 | 127.5±4.6 |
| $VO_2$ (mL・$kg^{-1}$・$min^{-1}$) | 3.7 ± 0.3 | 54.5 ± 1.3 |
| RER | 0.85 ± 0.01 | 1.18 ± 0.02 |
| $SpO_2$ (%) | 96 ± 1 | 94 ± 1 |

Data are shown as mean ± SD. BMI: body mass index; HR: heart rate; $V_E$: minute ventilation; $VO_2$: oxygen uptake; RER: respiratory exchange rate; $SPO_2$: saturation of peripheral oxygen.

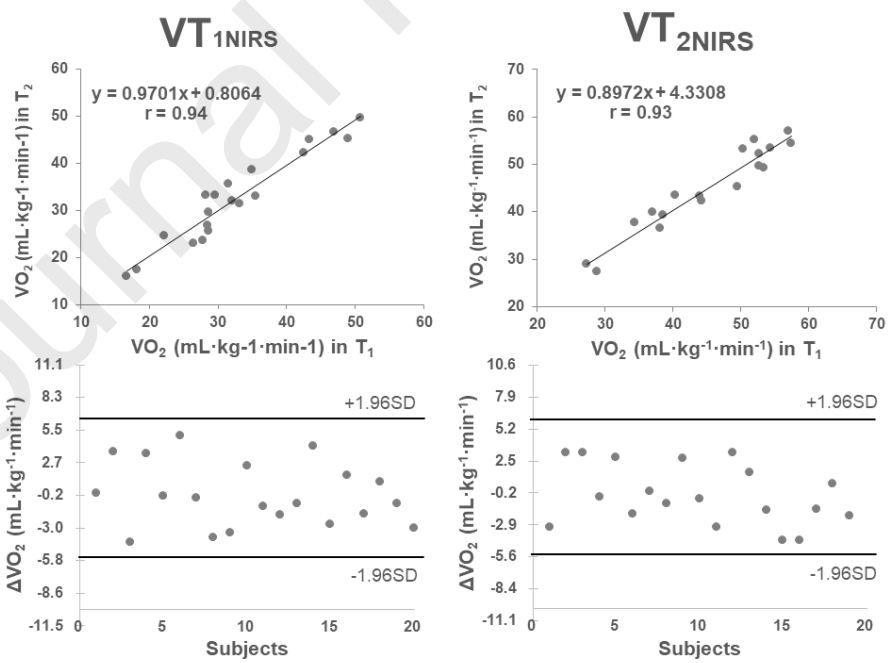


Figure 2. Reliability analysis between the first ($T_1$) and second ($T_2$) tests (n = 20).

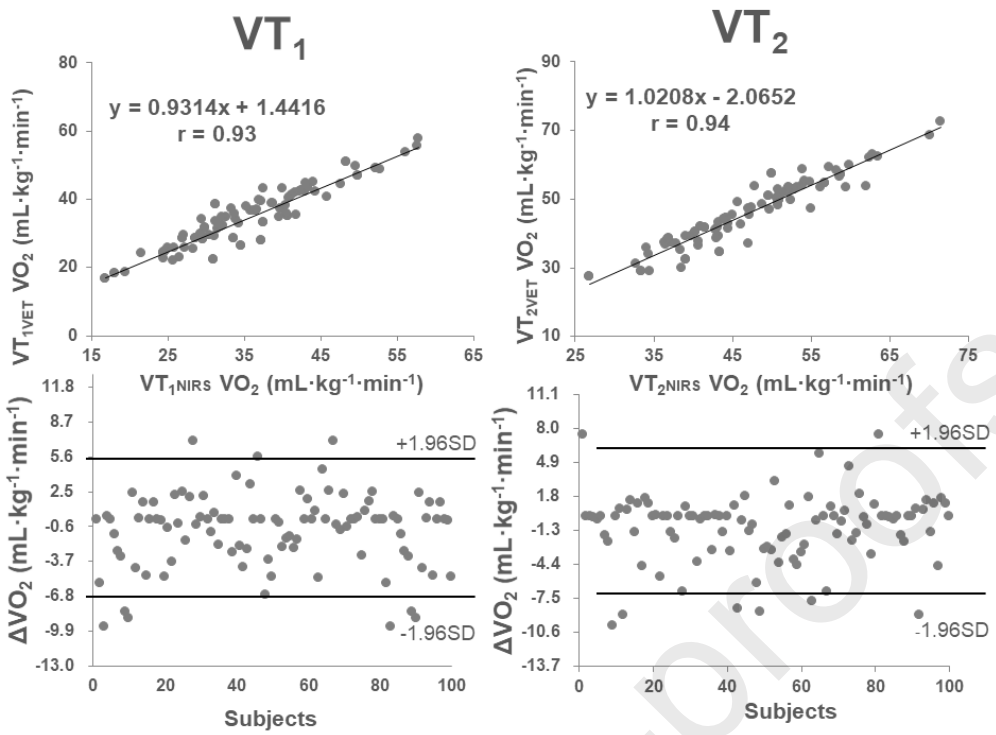


Figure 3. Reliability analysis of the ventilatory and NIRS methods (n = 100).

Table 2. Comparison of the cardiopulmonary parameters using the ventilatory and NIRS methods

| Parameters | $VT_{1VET}$ | $VT_{1NIRS}$ | $VT_{2VET}$ | $VT_{2NIRS}$ |
|---|---|---|---|---|
| Workload (watts) | 129 ± 3 | 122 ± 4 | 167 ± 3 | 159 ± 3 |
| HR (beat・$min^{-1}$) | 146 ± 1 | 144 ± 2 | 170 ± 1 | 165 ± 2 |
| $V_E$ (L・$min^{-1}$) | 60.9 ± 2.1 | 59.5 ± 2.2 | 93.0 ± 2.4 | 87.2 ± 2.2 |
| $VO_2$ (mL・$kg^{-1}$・$min^{-1}$) | 35.8 ± 1.0 | 35.0 ± 1.0 | 47.4 ± 1.0 | 46.2 ± 1.1 |
| RER | 1.01 ± 0.01 | 1.00 ± 0.01 | 1.09 ± 0.01 | 1.07 ± 0.01 |
| $SpO_2$ (%) | 96.1 ± 0.1 | 96.1 ± 0.1 | 96.1 ± 0.1 | 95.9 ± 0.2 |

Data are shown as mean ± SD. *$p < .05$ calculated using the paired t-test to compare the ventilatory and NIRS methods. NIRS: near-infrared spectroscopy; $VT_{1VET}$: first ventilatory threshold determined by ventilatory methods; $VT_{2VET}$: second ventilatory threshold determined by ventilatory methods; $VT_{1NIRS}$: first ventilatory threshold determined by NIRS; $VT_{2NIRS}$: second ventilatory threshold determined by NIRS; HR: heart rate; $VO_2$: oxygen uptake; $V_E$: minute ventilation; RER: respiratory exchange rate; $S_PO_2$: saturation of peripheral oxygen.

**4. Discussion**

This study's main findings are: 1) skeletal muscle oxygenated and deoxygenated status detected via NIRS are reliable and valid for determining $VT_1$ and $VT_2$ using a mathematical polynomial regression model, 2) $VT_{1NIRS}$ reflected the limitation of muscle oxygenation during the GXT, and 3) $VT_{2NIRS}$ reflected the limitation of muscle deoxygenation during the GXT.

NIRS is a valid method that can be used to determine VT during exercise as previously confirmed in many populations, including normal subjects, children (Moalla *et al.*, 2005), patients (Hamaoka *et al.*, 2007), and athletes (van der Zwaard *et al.*, 2016). NIRS was also confirmed as a non-invasive method of determining exercise intensity. Many studies demonstrated medium to high correlations between NIRS and traditional methods. The NIRS index demonstrated a decrease in oxyhemoglobin below baseline, a first sharp decrease in the oxygenation index, and first and second inflection early in the oxyhemoglobin slope (Bhambhani *et al.*, 1997; Miura *et al.*, 1998; Grassi *et al.*, 1999; Terakado *et al.*, 1999; Wang *et al.*, 2006; Soller *et al.*, 2008).

Most studies determined the NIRS index via intuitive or simple linear regression. Recent studies used more complex methods of determining the NIRS index, including tissue oxygen saturation using the Dmax method (Karatzanos *et al.*, 2010), non-linear changes in cerebral oxygenation or deoxygenation (Racinais *et al.*, 2014), and $\Delta O_2HbMb$-HHbMb by double linear regression (van der Zwaard *et al.*, 2016). The present study used NIRS origin muscle oxygenation ($O_2Hb$) and deoxygenation (HHb) data to determine VT and RCP using a polynomial regression model. $VT_{NIRS}$ and $RCP_{NIRS}$ in this study were respectively 59.4% and 86.1%$VO_{2max}$. Moalla et al. (2005)

investigated $VT_{NIRS}$ using respiratory muscle oxygenation data during an incremental exercise cycling test. $VT_{NIRS}$ was 73.8 ± 6.9% and $VT_{V\text{-}slope}$ was 69.7 ± 6.8% of $VO_{2max}$ (Moalla *et al.*, 2005). Karatzanos (2010) studied gastrocnemius muscle oxygen saturation during an incremental treadmill test. VT was 86.2 ± 6.9% and NT (NIRS threshold) was 88.7 ± 7.3% $VO_{2max}$ (Karatzanos *et al.*, 2010). NT was determined using a linear model. Racinais (2014) detected VL muscle oxygenation during an incremental cycling test. Cerebral $O_2Hb$ was 56 ± 8% and HHb was 56 ± 13% at VT1. Cerebral $O_2Hb$ was 86 ± 8% and HHb was 87 ± 8% at $VT_2$. Muscle oxygenation only related to VT2 of $O_2Hb$ was 78 ± 9% and HHb was 80 ± 8% (Racinais *et al.*, 2014). Zwaard et al. (2016) used maximal incremental cycling to detect oxygenation breakpoints via double linear regression (the least combined residual sum of squares). $\Delta O_2HbMb$-HHbMb was 65 ± 11 $\%PO_{peak}$ in male cyclists, 62 ± 12% $\%PO_{peak}$ in female cyclists, 58 ± 12 $\%PO_{peak}$ in endurance trained males, and 62 ± 10 $\%PO_{peak}$ in recreationally trained males. $\Delta O_2HbMb$-HHbMb was not significantly different from $VT_1$ but considerably lower than $VT_2$ (van der Zwaard *et al.*, 2016).

Our results were very similar to those using a non-linear regression model. This study also confirmed that $VT_{1VET}$ and $VT_{1NIRS}$ had a good correlation and $VT_{NIR}$ (59.4%) occurred earlier than VT (61.3%). As the exercise intensity increased, the oxygen deficit increased accompanied by anaerobic metabolism. Lactic acid in the blood began to accumulate and bicarbonate ions inside the muscle cells increased. Increasing intracellular carbon dioxide also led to an increase in $CO_2$ in the blood, which was reflected by changing ventilation. The initiation of anaerobic metabolism detected by ventilation was defined as VT (Wasserman *et al.*, 1981; Beaver *et al.*, 1986). Our results showed that decreased muscle oxygenation ($O_2Hb$) was a good

predictor of VT. Muscle oxygenation ($O_2Hb$) and deoxygenation (HHb) changed according to the oxyhemoglobin dissociation curve. Lactic acid, carbon dioxide, and hydrogen ions affected the oxyhemoglobin dissociation curve and increased the dissociation of oxygen from hemoglobin. A right shift of the Bohr effect indicated an decrease in $O_2Hb$ and a decrease in HHb in the muscle. Although $VO_2$ of $VT_{VET}$ and $VT_{NIRS}$ almost coincided, $VT_{NIRS}$ tended to occur earlier than $VT_{VET}$. Our findings confirmed that $VT_{NIRS}$ determined using raw data with a polynomial regression is a good predictor of $VT_{VET}$.

This study also confirmed that $VT_{2VET}$ and $VT_{2NIRS}$ have good correlation. $VT_{2NIRS}$ (86.1%) occurred earlier than $VT_{2VET}$ (88.6%). The definition of $VT_2$ was the starting point of respiratory compensation in response to exercise-induced metabolic acidosis. Beyond $VT_2$, the increase in hydrogen ions enhanced a shift to the right of the oxyhemoglobin dissociation curve and further increased oxygen dissociation. In this study, $VT_{2NIRS}$ was determined by the peak of HHb function, which may have been caused by the dissociation of oxygen from myoglobin. $VT_{2NIRS}$ also formed due to attenuated $O_2Hb$ decrease. The decrease in $O_2Hb$ might have increased oxygen dissociation or decreased blood flow. Our non-published data indicated that cardiac output declined during the highest intensity exercise, which might explain why the muscle blood flow also decreased as shown in the present results.

## 5. Conclusions

In conclusion, NIRS muscle oxygenation and deoxygenation are reliable and valid methods of determining $VT_1$ and $VT_2$ using a mathematical polynomial function model.

$VT_{1NIRS}$ reflected the limitation of muscle $O_2$ delivery ($O_2Hb$) and $VT_{2NIRS}$ reflected the limitation of muscle $O_2$ utilization (HHb) during incremental cycling exercise.

**6. Acknowledgments**

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